EXPERIMENT 20

Implement dilation as a morphological operation on foreground regions using OpenCV.

Aim

Implement dilation as a morphological operation on foreground regions using OpenCV.

Algorithm

• Import OpenCV and NumPy.

• Read the image and convert it to grayscale.

• Threshold the image to binary form.

• Create a kernel and apply morphological dilation.

• Save the dilated result.

Code

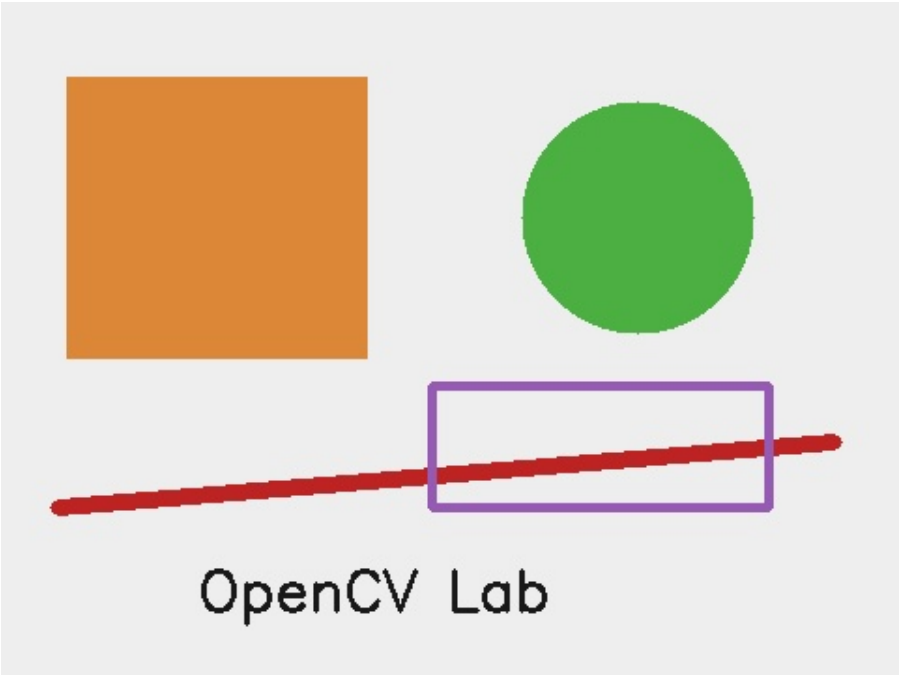
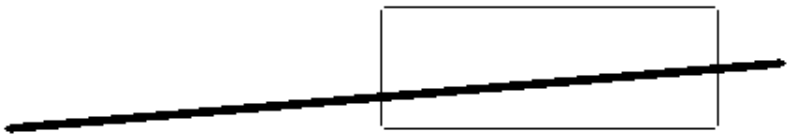
```
import cv2
import numpy as np

img = cv2.imread("images/input.jpg")
if img is None:
    raise FileNotFoundError("images/input.jpg not found")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
_, binary = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)
result = cv2.morphologyEx(binary, cv2.MORPH_DILATE, np.ones((5, 5), np.uint8))
cv2.imwrite("output_images/experiment20_morphological_dilation.png", result)
print("Saved output_images/experiment20_morphological_dilation.png")
```

Input Image:

Output Image:

Note: Morphological operations use a thresholded binary version of the sample image.

Input Image



Output Image